

# SARVESH PRAJAPATI

✉ [prajapati.s@northeastern.edu](mailto:prajapati.s@northeastern.edu)    [Sarvesh Prajapati](#)    [prajapatisarvesh](#)    [prajapatisarvesh](#)

## EDUCATION

- PhD in Electrical Engineering and Computer Engineering**, Northeastern University Dec 2028  
Focus: *Robotics*, and *Autonomous Systems*.
- M.S. in Robotics**, Northeastern University, GPA: 3.75/4.00 Dec 2024  
Thesis: *Physics-Informed Motion Planning Using RGB-Based Spectral Analysis*.  
Focus: *Algorithms*, *Reinforcement Learning*, *Robotic Systems*, *Controls*, *Manipulation*, and *Perception*.
- B.E. in Computer Engineering**, Gujarat Technological University, GPA: 3.68/4.00 Jun 2022  
Thesis: *Advancements in autonomous mobile robots for warehouse management and small scale industry*.  
Focus: *Robotics*, *DBMS*, *Data Structure and Algorithms*, *Operating Systems*, and *Object Oriented Design*.

## PUBLICATIONS

### Journal Papers

1. A. Trivedi, **S. Prajapati**, M. Zolotas, M. Everett, T. Padir, “Chance-Constrained Convex MPC for Robust Quadruped Locomotion Under Parametric and Additive Uncertainties,” *IEEE Robotics and Automation Letters (RA-L)*, 2025. **Best Paper Finalist (Top 3), IEEE RAS Technical Committee on Optimization for Robotics.**

### Under Review

1. **S. Prajapati**, A. Trivedi <sup>\*</sup>, L. M. Genua <sup>\*</sup>, D. Moore, T. Padir, “Universal Navigation Interface: Robot-Free Data for Wheeled Robot Navigation,” under review.
2. A. Trivedi, **S. Prajapati**, Z. Xu, M. K. M. Jaffar, D. Rosen, T. Padir, “Accelerating Optimization over Graphs of Convex Sets via Neural Network Approximations,” under review.

### Peer-Reviewed Conference Papers

1. L. M. Genua, **S. Prajapati**, D. Leblebicioglu, T. Padir, “Layered Risk Mapping for Autonomous Patient Transport in Expeditionary Medical Facilities,” 2026 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR), 2026.
2. **S. Prajapati**, A. Trivedi, N. Hanson, B. Maxwell, T. Padir, “Spectral Signature Mapping from RGB Imagery for Terrain-Aware Navigation,” IEEE International Conference on Robotic Computing (IRC), Naples, Italy, 2025. **Best Paper Award.**
3. A. Trivedi, **S. Prajapati**, A. Shirgaonkar, M. Zolotas, T. Padir, “Data-Driven Sampling Based Stochastic MPC for Skid-Steer Mobile Robot Navigation,” 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, USA, 2025.
4. N. U. Akmandor, **S. Prajapati** <sup>\*</sup>, M. Zolotas <sup>\*</sup>, T. Padir, “Re4MPC: Reactive Nonlinear MPC for Multi-model Motion Planning via Deep Reinforcement Learning,” 2025 IEEE International Conference on Automation Science and Engineering (CASE), Los Angeles, CA, USA, 2025.
5. N. Hanson <sup>\*</sup>, **S. Prajapati** <sup>\*</sup>, J. Tukpah, Y. Mewada, and T. Padir, “Automated Forest Biomass Mapping with Autonomous Hyperspectral Imaging for Wildfire Monitoring,” 2024 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR), New York, US, 2024.
6. A. Trivedi, M. Zolotas, A. Abbas, **S. Prajapati**, S. Bazzi, and T. Padir, “A Probabilistic Motion Model for Skid-Steer Wheeled Mobile Robot Navigation on Off-Road Terrains,” 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, JP, 2024.

<sup>\*</sup>equal contribution

7. Y. Qian, **S. Prajapati**, A. Schwartz, A. Jung, U. Seitz, J. Alfen, L. Lewis, M. Kim, A. Kramer, L. Chukoskie, "Integrated Aerobic Exercise into Adult Second Language Learning in Virtual Reality Game," 2023 IEEE Conference on Games (CoG), Boston, MA, USA, 2023.

### Workshop Papers

1. **S. Prajapati**, A. Trivedi \*, L. M. Genua \*, D. Moore, T. Padir, "Universal Navigation Interface: Robot-Free Data for Wheeled Robot Navigation," Workshop on Data in Field Robotics: From State Estimation to Navigation, IROS, Pittsburgh, PA, USA, 2026. **Best Student Paper Award.**
2. L. M. Genua, **S. Prajapati**, A. Trivedi, T. Padir, "Shades of Success: Characterizing Manipulation Quality through Object-Centric Constraints," Workshop on Geometric Representations in Robotics, IROS, Pittsburgh, PA, USA, 2026. **Best Paper Award.**
3. **S. Prajapati**, A. Trivedi, B. Maxwell, and T. Padir, "Predictive Mapping of Spectral Signatures from RGB Imagery for Off-Road Terrain Analysis," Workshop on Resilient Off-road Autonomy, ICRA, Yokohama, JP, 2024.
4. A. Trivedi, **S. Prajapati**, M. Zolotas, T. Padir, "Online Refinement of Uncertainty Sets for Robust MPC of Quadrupedal Robots Using Convex Cone Programming," Workshop on Advancements in Trajectory Optimization and Model Predictive Control, ICRA, Yokohama, JP, 2024.

### RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Research Assistant</b><br><b>RIVeR Lab</b> , Northeastern University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jan 2023 – <i>Present</i><br>Boston, US |
| <ul style="list-style-type: none"> <li>• Developing sidewalk navigation for food-delivery and autonomous wheelchair robots.</li> <li>• Researching on deep learning-based methods to map RGB to hyperspectral images for material recognition and friction estimation enhancing the traversability for mobile and legged robots.</li> <li>• Investigated control systems, perception, and learning algorithms, contributing to the design of autonomous systems.</li> <li>• Mentoring and managing undergraduate, graduate, and co-op students on lab research projects.</li> </ul> |                                         |
| <b>Applied Scientist Co-Op</b><br><b>Amazon Robotics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Feb 2025 – Dec 2025<br>Westborough, US  |
| <ul style="list-style-type: none"> <li>• Researching in Innovation Lab, NDA bound.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |
| <b>Applied Scientist Co-Op</b><br><b>Amazon Robotics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Jan 2024 – Jun 2024<br>Westborough, US  |
| <ul style="list-style-type: none"> <li>• Researched in Innovation Lab, NDA bound; work involved building simulation environments and integrating motion planning, perception, and LLM-based methods for robotic manipulators.</li> </ul>                                                                                                                                                                                                                                                                                                                                            |                                         |
| <b>Research Assistant</b><br><b>ReGame-XR Lab</b> , Northeastern University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sep 2022 – Dec 2022<br>Boston, US       |
| <ul style="list-style-type: none"> <li>• Researched on XR game development and cognition for secondary language learning in older adults.</li> <li>• Led software development team for <b>ExerBike project</b>; modified GTA V, interfaced ANT+ sensor, integrated VR using .NET framework that helped securing \$50k CBH award and conference publication.</li> </ul>                                                                                                                                                                                                              |                                         |
| <b>Research Scholar</b><br><b>Gujarat Technological University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jun 2020 – Aug 2022<br>Ahmedabad, IN    |
| <ul style="list-style-type: none"> <li>• Researched on controls, embedded systems, motion generation and path-planning for mobile robots. Migrated robots controller from Arduinos to STM32 and wrote a primer on STM32 for undergrads.</li> </ul>                                                                                                                                                                                                                                                                                                                                  |                                         |

### TEACHING EXPERIENCE

|                                                                                                                                                                                                                                                   |                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Advanced Perception</b> , Northeastern University<br>Teaching Assistant                                                                                                                                                                        | Sep 2024 – Dec 2024<br>Boston, US |
| <ul style="list-style-type: none"> <li>• Responsible for grading students' weekly research paper reviews and presentations, offering office hours for guidance, and assisting in developing novel aspects for research-based projects.</li> </ul> |                                   |

**Embedded Design Enabling Robotics**, Northeastern University

Sep 2023 – Dec 2023

Teaching Assistant

Boston, US

- Conducted lab sessions for 100+ students using the De1SoC FPGA board, guiding them through assignments from basic logic gate design to programming a robotic arm.

**GTU Robotics Club**, Gujarat Technological University

Jul 2021 – Jul 2022

Team Leader

Ahmedabad, IN

- Taught 40 peers robot programming, computer vision and machine learning over a course of 1 year.

---

**HONORS AND AWARDS**

|      |                                                                                   |               |
|------|-----------------------------------------------------------------------------------|---------------|
| 2023 | <b>CS Research Mentorship Program Scholar</b> , Google                            | Boston, US    |
| 2022 | <b>Robotics team received best design award and \$1000 prize</b> , DD Robocon     | Delhi, IN     |
| 2021 | <b>Team represented India in international robotics competition</b> , ABU Robocon | Jimo, CN      |
| 2021 | <b>Code-Decode champion</b> , Parul University                                    | Vadodara, IN  |
| 2020 | <b>Won CTF and OSCP voucher</b> , Secarmy                                         | Ahmedabad, IN |

---

**SERVICE AND VOLUNTEERING**

1. Reviewer for IEEE Robotics and Automation Letters (RA-L), IEEE conferences (ICRA, IROS, SSRR), ICML, and Amazon internal conferences.
2. Mentored undergraduates, guided them in robotics research and projects.

---

**SKILLS**

|                              |   |                                                                                                                                               |
|------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | : | C++, Python, MATLAB, JavaScript, Bash                                                                                                         |
| <b>Software/Tools</b>        | : | ROS 2, Linux, CMake, Docker, MoveIt!, Git, Gazebo, Nvidia Isaac Sim, $\LaTeX$                                                                 |
| <b>Libraries</b>             | : | PyTorch, JAX, OpenCV, PCL, Open3D, Drake, CasADi, CVXPY                                                                                       |
| <b>Platforms</b>             | : | Clearpath, Universal Robots, Franka Emika, Unitree, Human Support Robot, Xarm, Kinova, TurtleBot, Nvidia Jetson, Raspberry Pi, Arduino, STM32 |